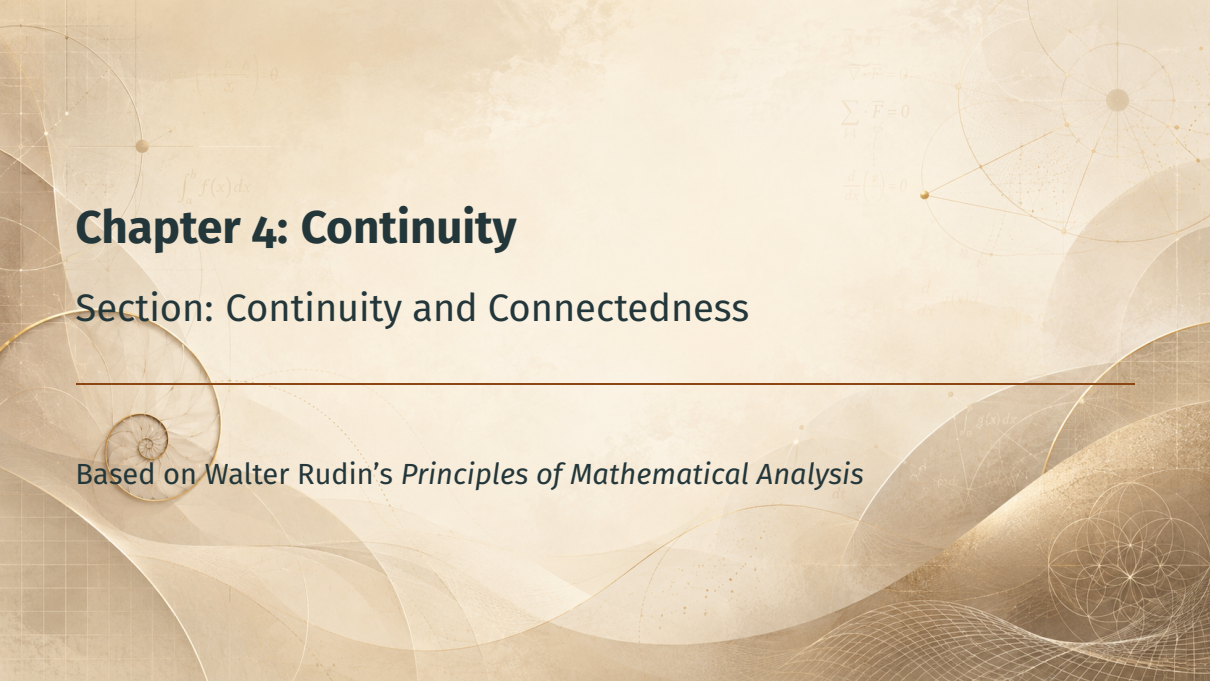


The background is a light beige color with various mathematical and geometric elements. In the top left, there is a partial view of a golden spiral. In the top right, there are faint, overlapping circles and lines, resembling a geometric construction or a network diagram. Scattered throughout the background are several mathematical formulas in a light, semi-transparent font. These include: $\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right) \cdot \vec{F}$ in the top left; $\int_a^b f(x) dx$ in the middle left; $\vec{\nabla} \cdot \vec{F} = 0$ in the top right; $\sum_{i=1}^n \vec{F}_i = 0$ in the middle right; $\frac{d}{dx} \left(\frac{g}{f} \right) = 0$ in the bottom right; $\frac{d}{dt} g(x) dx$ in the bottom right; and $\frac{d}{dt} g(x) dx$ in the bottom right. The main title 'Chapter 4: Continuity' is in a bold, dark blue font, and the subtitle 'Section: Continuity and Connectedness' is in a regular, dark blue font. A thin horizontal line is positioned below the subtitle. The text 'Based on Walter Rudin's Principles of Mathematical Analysis' is in a regular, dark blue font at the bottom left.

Chapter 4: Continuity

Section: Continuity and Connectedness

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

The Continuous Image of a Connected Set

The Intermediate Value Theorem

Is There a Converse?

Summary

Motivation

Why This Section Matters

- Continuity preserves **compactness** (last section)

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- Does continuity also preserve **connectedness**?

Why This Section Matters

- Continuity preserves **compactness** (last section)
- Does continuity also preserve **connectedness**?
- Payoff: the **Intermediate Value Theorem**

Recall: Connected Sets (Chapter 2)

Definition 2.45: Separated and Connected

Sets $A, B \subset Y$ are **separated** if

$$A \cap \bar{B} = \emptyset \quad \text{and} \quad \bar{A} \cap B = \emptyset.$$

A set is **connected** if it is *not* a union of two nonempty separated sets.

Key idea: separated is stronger than disjoint — neither set touches the other's *closure*.

$$\nabla \cdot F = 0$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\int_a^b f(x) dx$$

$$e^{i\pi} + 1 = 0$$

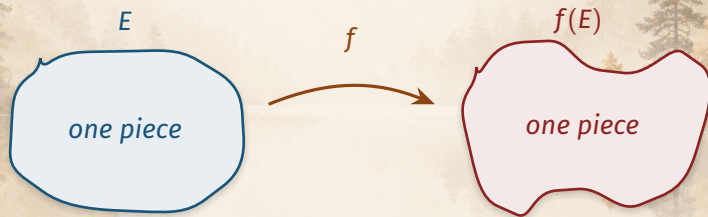
The Continuous Image of a Connected Set

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Theorem 4.22: Connectedness Is Preserved

Theorem 4.22

If f is a continuous mapping of a metric space X into a metric space Y , and if $E \subset X$ is **connected**, then $f(E)$ is **connected**.



Proof Strategy: Theorem 4.22

Technique: proof by contradiction.

1. Suppose $f(E)$ is split: $f(E) = A \cup B$, separated and nonempty.
2. **Pull back** the split to the domain: $G, H \subset E$.
3. Show G and H are nonempty with $E = G \cup H$.
4. Use continuity to show G and H are **separated**.
5. This contradicts E being connected. $\Rightarrow \Leftarrow$

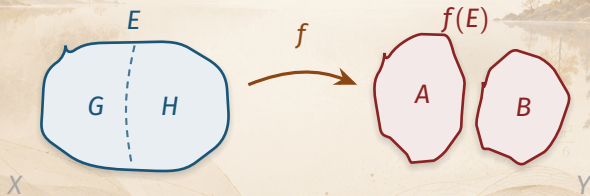
Proof of 4.22 — Step 1: Assume a Split

Step 1: Suppose the image is disconnected

Assume $f(E) = A \cup B$, where A, B are nonempty **separated** subsets of Y .

$$G = E \cap f^{-1}(A), \quad H = E \cap f^{-1}(B).$$

Idea: G is the part of E landing in A ; H the part landing in B .



Proof of 4.22 — Step 2: They Cover E

Step 2: $E = G \cup H$, both nonempty

Since $f(E) = A \cup B$, every point of E maps into A or B :

$$E = G \cup H.$$

As A, B are nonempty, so are G and H .

Proof of 4.22 — Step 3: Continuity Closes G Down

Step 3: $f(\bar{G}) \subset \bar{A}$

$A \subset \bar{A} \Rightarrow G \subset f^{-1}(\bar{A})$. Since f is continuous and $\bar{A}$ is closed, $f^{-1}(\bar{A})$ is **closed**, so

$$\bar{G} \subset f^{-1}(\bar{A}) \implies f(\bar{G}) \subset \bar{A}.$$

Proof of 4.22 — Step 4: $\bar{G}$ and H Don't Touch

Step 4: $\bar{G} \cap H = \emptyset$

$f(H) = B$ and $\bar{A} \cap B = \emptyset$. Since $f(\bar{G}) \subset \bar{A}$:

$$\bar{G} \cap H = \emptyset.$$

By symmetry, $G \cap \bar{H} = \emptyset$ as well.

Hence G and H are **separated**.

Proof of 4.22 — Conclusion

Contradiction

$E = G \cup H$, with G, H nonempty and **separated**.

This is impossible if E is connected.

So no such split $A \cup B$ exists: $f(E)$ is connected. ■

$$e^{i\pi} + 1 = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$y = \frac{\sin x}{x}$$

The Intermediate Value Theorem

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

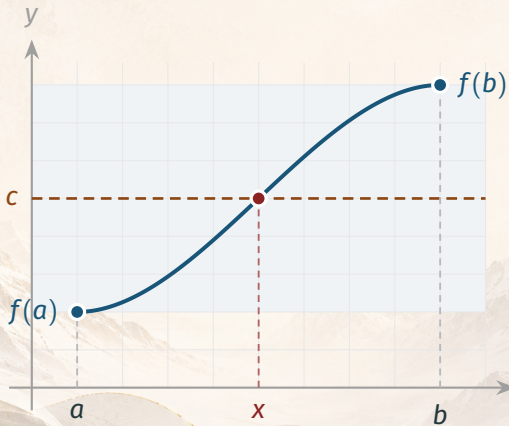
Theorem 4.23: Intermediate Value Theorem

Theorem 4.23

Let f be a continuous real function on $[a, b]$. If $f(a) < f(b)$ and $f(a) < c < f(b)$, then there exists $x \in (a, b)$ with $f(x) = c$.

A continuous real function on an interval **assumes every intermediate value**. (Same conclusion if $f(a) > f(b)$.)

Intermediate Value Theorem — Picture



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$y = \frac{\sin x}{x}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Proof of 4.23: Two Appeals to Theorem 2.47

Proof

1. By **Theorem 2.47**, $[a, b]$ is connected.
2. By **Theorem 4.22**, $f([a, b])$ is a connected subset of $\mathbb{R}^1$.
3. By **Theorem 2.47** again, $f([a, b])$ has the *in-between property*; as it contains $f(a), f(b)$, it contains every c with $f(a) < c < f(b)$.

Recall: Theorem 2.47

$E \subset \mathbb{R}^1$ is connected $\iff$ if $x, y \in E$ and $x < z < y$, then $z \in E$.

The background is a complex, artistic composition. It features a landscape with rolling hills and mountains in the distance, rendered in a soft, painterly style. Overlaid on this are various mathematical and geometric elements. In the upper left, there are faint, golden grid lines and a lightning bolt. In the center, the title 'Is There a Converse?' is written in a bold, dark blue font, underlined with a thin brown line. To the left of the title, there is a golden spiral shell. To the right, there are several mathematical equations in a golden, serif font. The overall color palette is dominated by warm, golden-brown and light blue tones, creating a dreamlike and intellectual atmosphere.

Is There a Converse?

$$\frac{d^2 y}{dx^2} + \omega^2 y = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$\nabla^2 \phi = 0$$

$$e^{im} + 1 = 0$$

$$\frac{\partial u}{\partial t} = \sigma \nabla^2 u$$

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Remark 4.24: A Tempting Converse

Remark 4.24: does the converse hold?

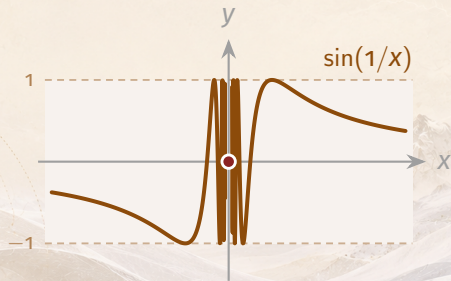
One might guess: if f takes every intermediate value on every subinterval, must f be **continuous**?

More precisely: for all $x_1 < x_2$ and every c between $f(x_1)$ and $f(x_2)$, there is $x \in (x_1, x_2)$ with $f(x) = c$.

Remark 4.24: The Converse Is False

Remark 4.24: does the converse hold?

One might guess: if f takes every intermediate value on every subinterval, must f be **continuous**?



No. Example 4.27(d): $f(x) = \sin(1/x)$, $f(0) = 0$.


$$e^{i\pi} + 1 = 0$$

$$\phi = \frac{1 + \sqrt{5}}{2}$$

Summary

$$\nabla^2 \psi + k^2 \psi = 0$$

$$\frac{d}{dx}(\sin x) = \cos x$$

Summary

Key takeaways:

- **Theorem 4.22:** continuity preserves *connectedness* — $f(E)$ is connected whenever E is.

$$\nabla^2 \psi + k^2 \psi = 0$$

$$\frac{d}{dx}(\sin x) = \cos x$$

Summary

Key takeaways:

- **Theorem 4.22:** continuity preserves *connectedness* — $f(E)$ is connected whenever E is.
- **Theorem 4.23:** the *Intermediate Value Theorem* — a continuous function on $[a, b]$ hits every value between $f(a)$ and $f(b)$.

Summary

Key takeaways:

- **Theorem 4.22:** continuity preserves *connectedness* — $f(E)$ is connected whenever E is.
- **Theorem 4.23:** the *Intermediate Value Theorem* — a continuous function on $[a, b]$ hits every value between $f(a)$ and $f(b)$.
- **Remark 4.24:** the converse *fails* — the intermediate value property does not imply continuity ($\sin(1/x)$).

Coming up next: *Discontinuities* — classifying the ways continuity can break down.